Files `correlators.mres, ad.mres` and `rc.mres` contain the MATHEMATICA versions (in general covariant gauge and for general gauge group) of the expressions given in Sections 6 and Appendixes A and B of the paper

V. M. Braun, K. G. Chetyrkin and B. Kniehl "Three-loop renormalization of parton quasi-distributions: spacelike vs. timelike".

Notations:

`xil` $= 1 -$ `xi`, `xi` $= \xi$: gauge parameter (`xil=0 ->` Landau gauge, `xil=1 ->` Feynman gauge); the tree gluon propagator reads:

$$D_{\alpha\beta}(q) = \frac{-g_{\alpha\beta}}{-q^2} + \xi \frac{q_\alpha q_\beta}{q^4}.$$

`z3, z5` stand for the values of the Riemann zeta-function with `z3` $= \zeta_3 = 1.2020569\ldots$, `z5` $= \zeta_5 = 1.0369278\ldots$.
`cf` $= C_F$ and `ca` $= C_A$ are the quadratic Casimir operators of the quark $[T^a T^a]_{ij} = C_F \delta_{ij}$ and the adjoint $[C^a\, C^a]_{bd} = C_A\, \delta_{bd}$, $(C^a)_{bc} = -i f^{abc}$ representations of the Lie algebra, `ng` and `dR` are dimensions of the adjoint (gluon) and quark representations of the gauge group, `nf` stands for the number of quark flavors and `tr` $= T_f$ refers to the trace normalization $\mathrm{tr}\,(T^a T^b) = T_f\, \delta^{ab}$.

Finally, `om` $= \omega = pv$, `vv` $= v \cdot v$ and

$$\texttt{tLn} = \ln \frac{\mu}{-2\omega},\ \texttt{sLn} = \ln \frac{\mu}{2\omega},\ \texttt{xLn} = \ln\left(\frac{\mu\, i\, t\, e^{\gamma_E}}{2}\right), \texttt{zLn} = \ln\left(\frac{\mu z\, e^{\gamma_E}}{2}\right),$$

with $\gamma_E = 0.577216\ldots$, being the Euler constant appearing in the expansion of the Gamma function

$$\Gamma(1+\epsilon) = 1 - \gamma_E\, \epsilon + \mathcal{O}(\epsilon^2).$$

File `correlators.mres` contains expressions
`tSigma, sSigma, tD, sD, tDtt, sDtt, tDpt, sDpt, zPitt, xPitt, zPipt, xPipt`
which correspond to the functions
$\Sigma(v^2=1)$, $\Sigma(v^2=-1)$, $tD(v^2=1)$, $\tilde{D}(v^2=-1)$, $\tilde{D}_{\perp\perp}(v^2=1)$, $\tilde{D}_{\perp\perp}(v^2=-1)$, $\tilde{D}_{\parallel\perp}(v^2=1)$, $\tilde{D}_{\parallel\perp}(v^2=-1)$, $\Pi_{\perp\perp}(z, v^2=-1)$, $\Pi_{\perp\perp}(t, v^2=1)$, $\Pi_{\parallel\perp}(z, v^2=-1)$, $\Pi_{\parallel\perp}(t, v^2=1)$.

Note, that have used a symbol `Im1` $\equiv i \equiv \sqrt{-1}$ istead of the Mathematica symbol `I` for the the imaginary unit. A symbol `x` is introduced as convenient label in the relation (see eq. (3.43) of the main text of the paper)

$$\texttt{tLn} \to \texttt{sLn} + \texttt{x} * (\texttt{Im1} * \texttt{Pi}/2). \tag{1}$$

Within a Mathematcia session it can always removed by saying

```
x = 1;
```

File `rc.mres` contains the renormalization constants `Za, Z2, Z3, ZQ, Z2h, Zpt,Ztt` (see Sections 3.1 and 3.2 of the paper).

File `ad.mres` contains the QCD β-function=`be`, as well as the corresponding ADs

`g2`=γ_2, `g3`=γ_3, `gQ`=γ_Q, `g2h`=γ_h, `gtt`=$\gamma_{\perp\perp}$, `gpt`=$\gamma_{\parallel\perp}$.

Bibliography notes:

Many results given in the paper and in the auxiliary files were originally obtained by other authors and have been included by us for completeness. Below we cite the corresponding works. Note that all results for ADs γ_h, γ_Q and for the VEV Π cited below correspond to the timelike choice of the vector v with $v \cdot v = 1$.

The anomalous dimension (AD) γ_Q was computed at 1- 2- and 3-loops in [1, 2], [3, 4] and [5] respectively.

The anomalous dimension of the special "heavy" field h_v was first computed at 2- and 3-loop in [4] and [6] and confirmed later in [5] and even extended to 4 loops in [7, 8].

The VEV Π was computed at 2 loops in [4] and at 3 loops in [9].

The VEV of gluon off-light-cone operator $\mathbb{G}_{\mu\nu\alpha\beta}$ in two loops was found in [10] and the corresponding 1-loop ADs first computed in [11, 12].

The 3-loop renormalization factors Z_3, Z_2 and Z_a are well-known since long [13, 14, 15].

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